```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/aerofit_treadmill.txt")
df.head()
```

| | Product | Age | Gender | Education | MaritalStatus | Usage | Fitness | Income | Miles |
|---|---|---|---|---|---|---|---|---|---|
| **0** | KP281 | 18 | Male | 14 | Single | 3 | 4 | 29562 | 112 |
| **1** | KP281 | 19 | Male | 15 | Single | 2 | 3 | 31836 | 75 |
| **2** | KP281 | 19 | Female | 14 | Partnered | 4 | 3 | 30699 | 66 |
| **3** | KP281 | 19 | Male | 12 | Single | 3 | 3 | 32973 | 85 |
| **4** | KP281 | 20 | Male | 13 | Partnered | 4 | 2 | 35247 | 47 |

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(180, 9)
```

```
df.columns
```

```
Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus', 'Usage',
       'Fitness', 'Income', 'Miles'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product        180 non-null    object
 1   Age            180 non-null    int64
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64
 6   Fitness        180 non-null    int64
 7   Income         180 non-null    int64
 8   Miles          180 non-null    int64
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

```
df["Product"].value_counts()
df["Gender"].value_counts()
df["MaritalStatus"].value_counts()
```

| MaritalStatus | count |
|---|---|
| **Partnered** | 107 |
| **Single** | 73 |

**dtype:** int64

```
df.nunique()
```

| | 0 |
|---|---|
| **Product** | 3 |
| **Age** | 32 |
| **Gender** | 2 |
| **Education** | 8 |
| **MaritalStatus** | 2 |
| **Usage** | 6 |
| **Fitness** | 5 |
| **Income** | 62 |
| **Miles** | 37 |

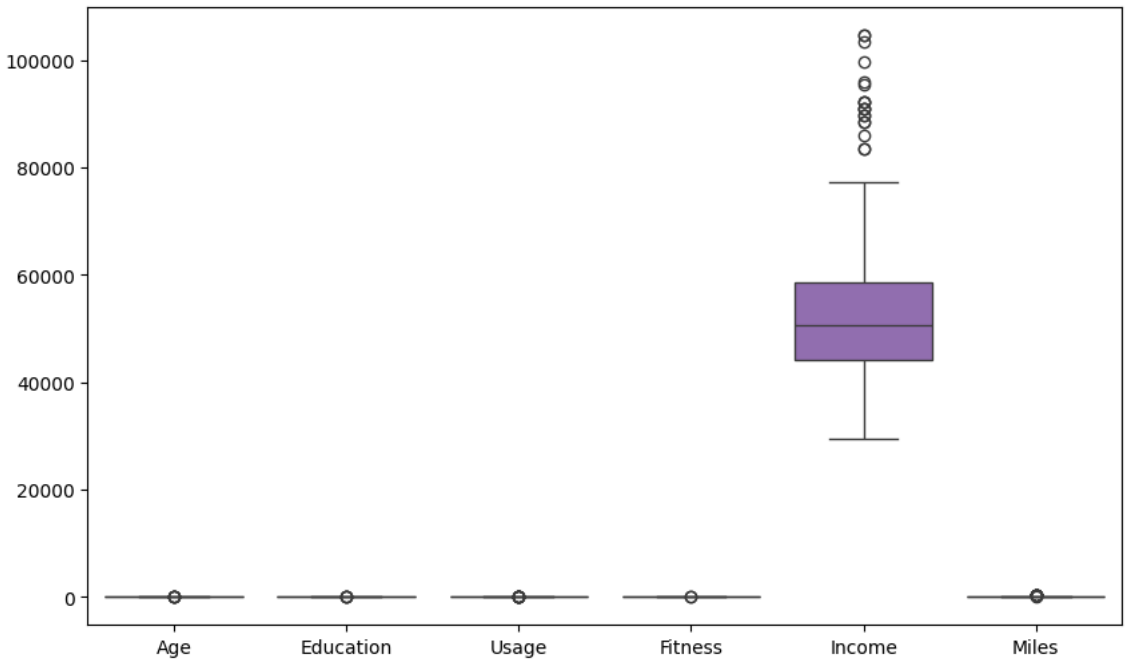
**dtype:** int64

```
df.isnull().sum()
```

| | 0 |
|---|---|
| **Product** | 0 |
| **Age** | 0 |
| **Gender** | 0 |
| **Education** | 0 |
| **MaritalStatus** | 0 |
| **Usage** | 0 |
| **Fitness** | 0 |
| **Income** | 0 |
| **Miles** | 0 |

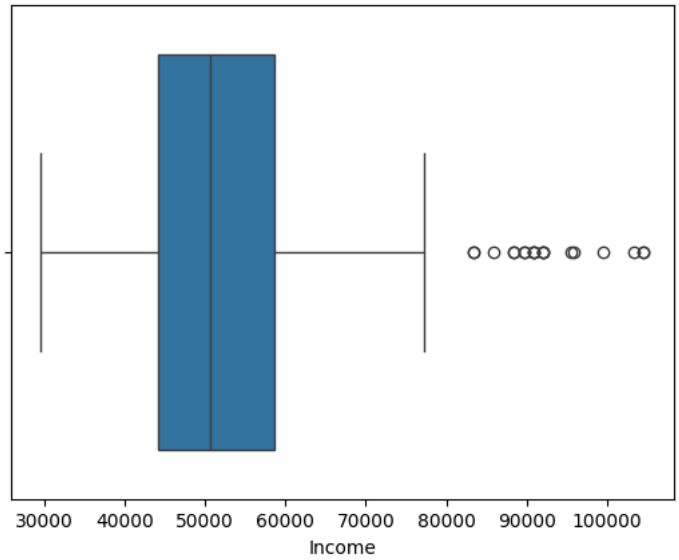
**dtype:** int64

```
plt.figure(figsize=(10,6))
sns.boxplot(data=df)
plt.show()
```

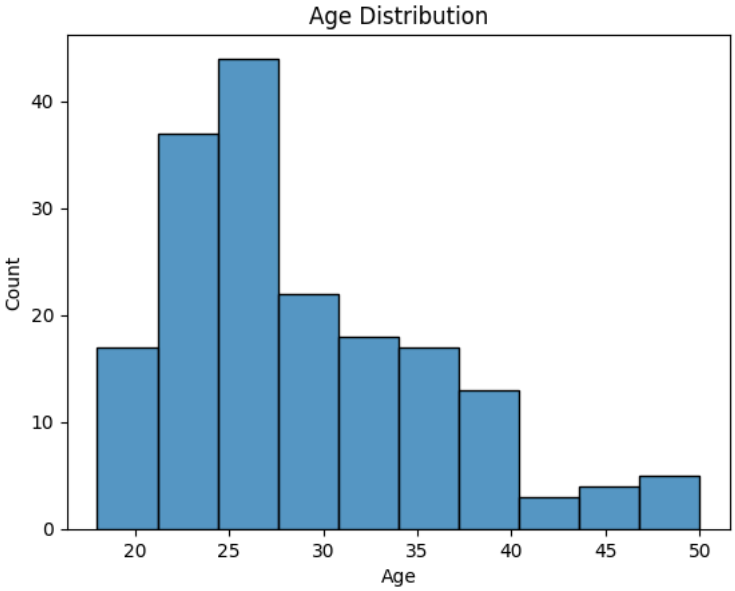



```
sns.boxplot(x=df["Income"])
```

```
<Axes: xlabel='Income'>
```



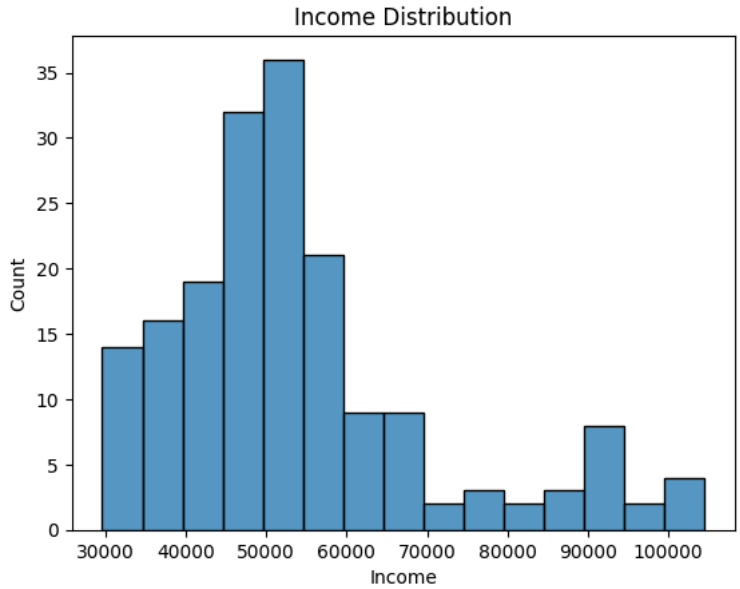

```
sns.histplot(df["Age"], bins=10)
plt.title("Age Distribution")
plt.show()
```




Most customers are 20-30

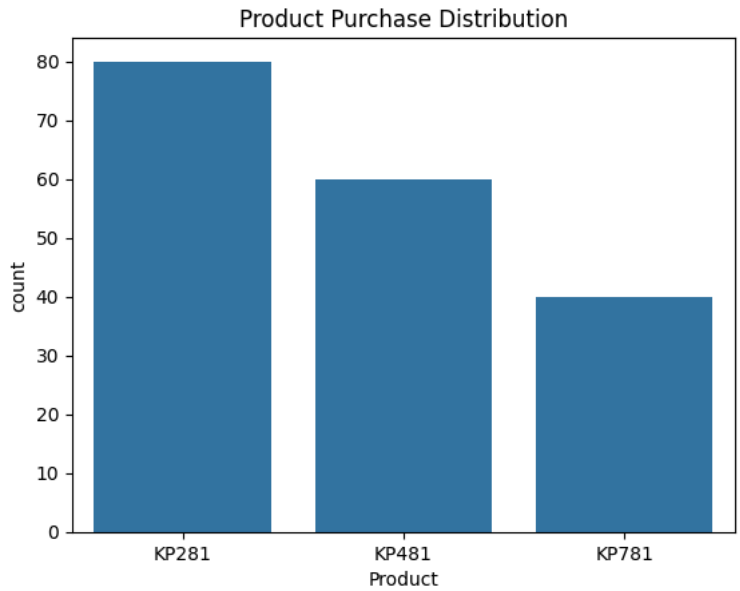
```
sns.histplot(df["Income"])
plt.title("Income Distribution")
```

```
Text(0.5, 1.0, 'Income Distribution')
```




```
sns.countplot(x="Product", data=df)
plt.title("Product Purchase Distribution")
```

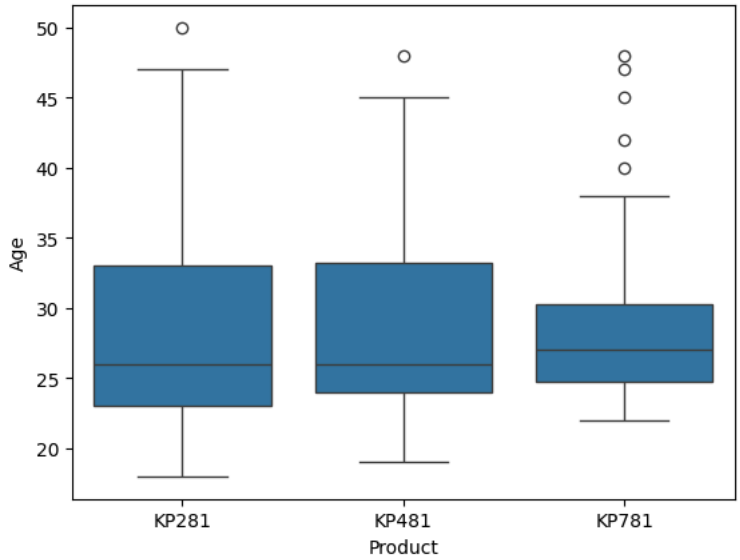
```
Text(0.5, 1.0, 'Product Purchase Distribution')
```




KP281 has the highest sales.

```
sns.boxplot(x="Product", y="Age", data=df)
```

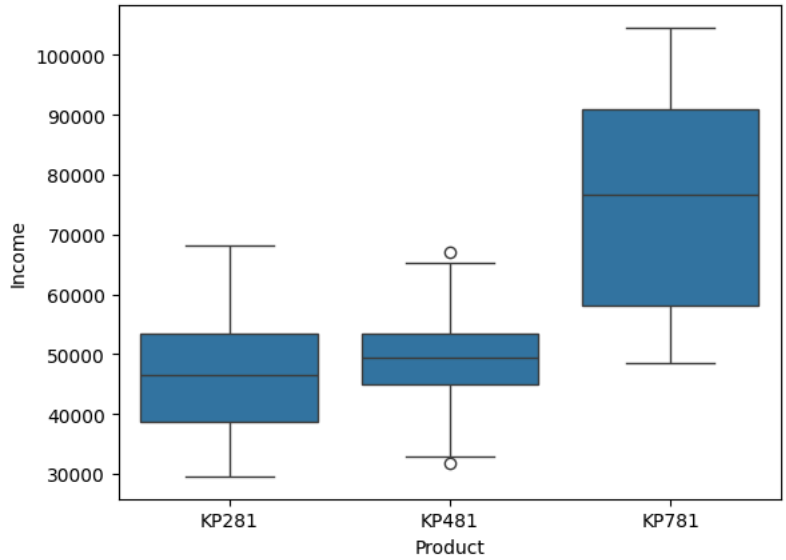
```
<Axes: xlabel='Product', ylabel='Age'>
```




KP781 buyers tend to be older customers.

```
sns.boxplot(x="Product", y="Income", data=df)
```

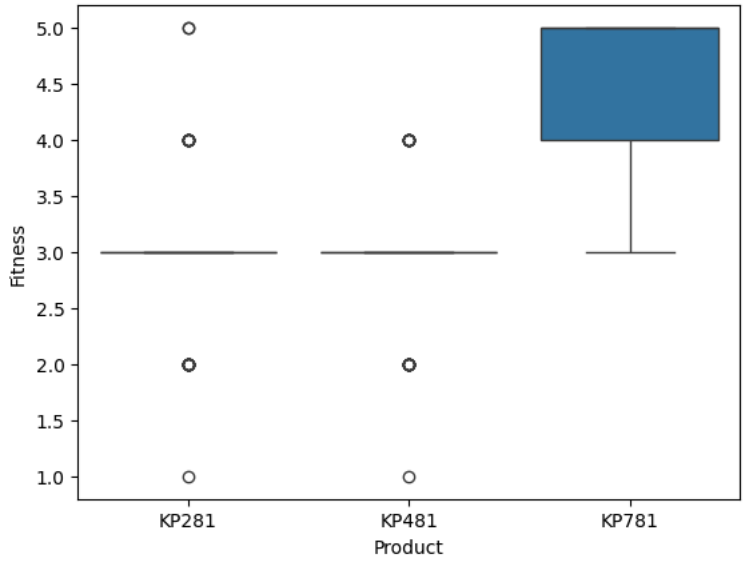
```
<Axes: xlabel='Product', ylabel='Income'>
```




KP781 buyers usually have higher income.

```
sns.boxplot(x="Product", y="Fitness", data=df)
```

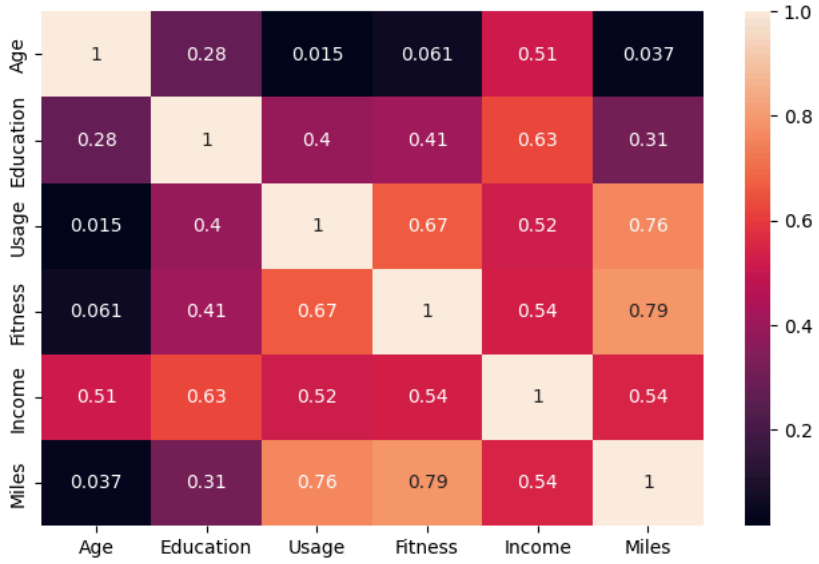
```
<Axes: xlabel='Product', ylabel='Fitness'>
```

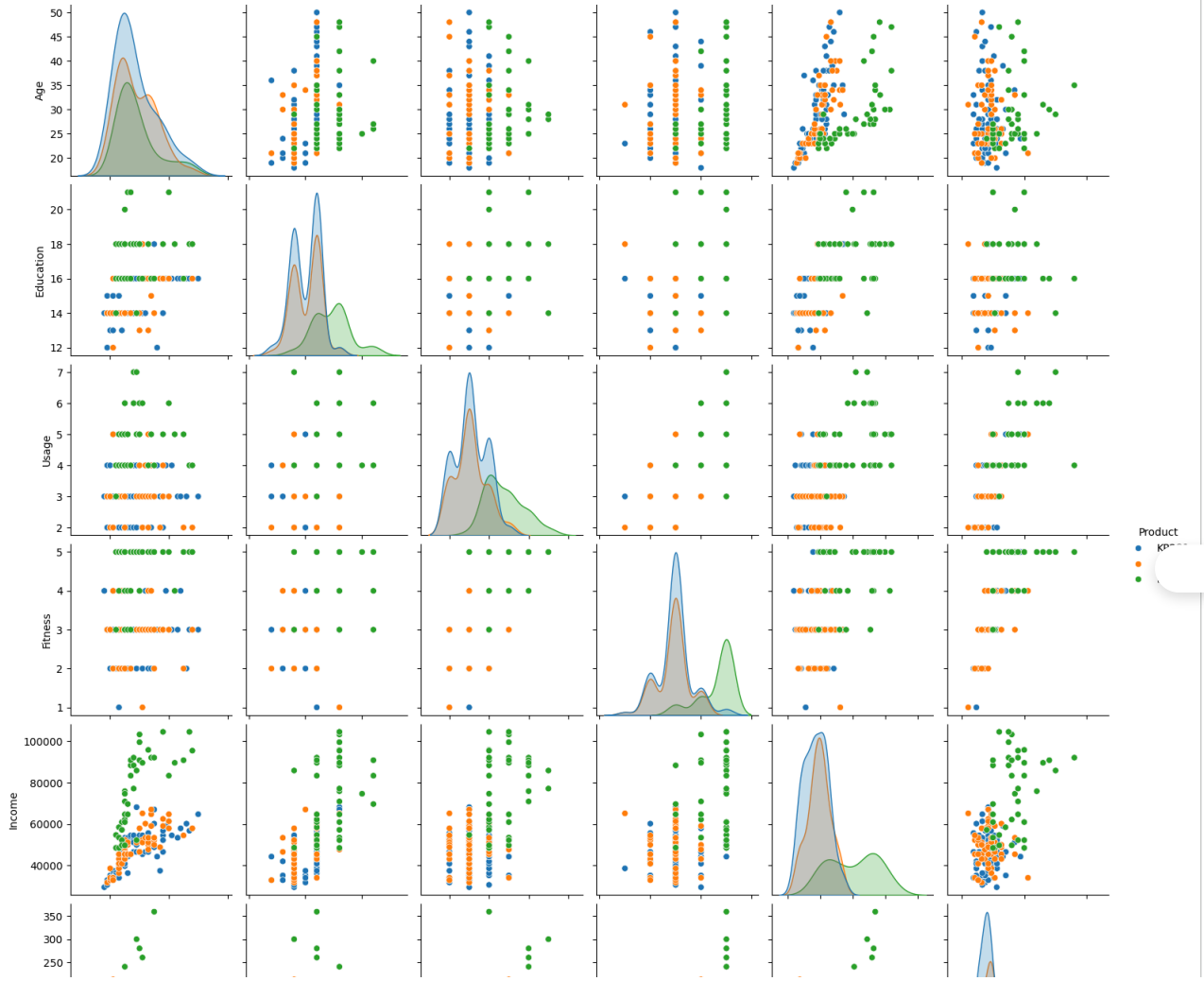



KP781 customers have higher fitness levels.

```
corr = df.corr(numeric_only=True)

plt.figure(figsize=(8,5))
sns.heatmap(corr, annot=True)
plt.show()
```




```
sns.pairplot(df, hue="Product")
```

```
<seaborn.axisgrid.PairGrid at 0x7b61b9750aa0>
```